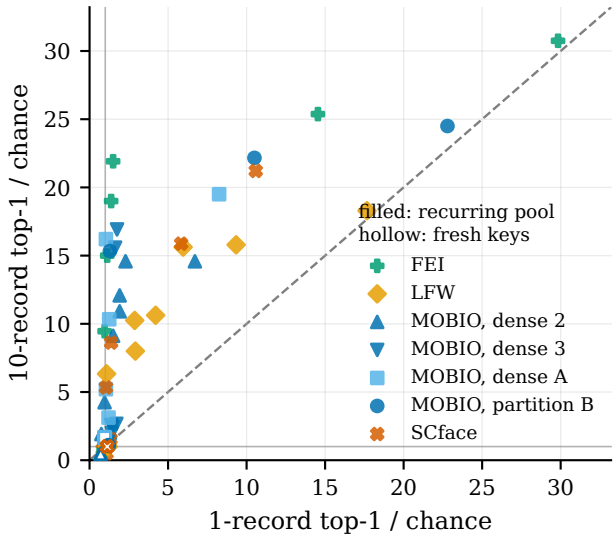


BioHash record-count gains



A DATA AND PROTECTION



1 Identity-disjoint data

MOBIO / LFW / FEI / SCface
Train / val / test people differ
Gallery never in exposures

2 Fixed face encoder

YuNet; 5-landmark alignment
ArcFace w600k_r50; 512-D
Unit norm; frozen backbone

3 Keyed protection

Template $T = f(\text{key}, x)$
Fresh / pool / shared key
Secret key values withheld

4 Exposure records

Input: batch $x \times n \times d$
8 nested sets per identity
 $n = 1, 10$ (initial: 1, 2, 5, 10)

OUTPUT d BioHash: 128 bits | MLP-Hash: 512 bits | IoM-GRP: 300 codes ($q=16$; one-hot $d=4,800$)
PolyProtect: 170 reals (window $m=5$, overlap=2). Scheme coverage differs by dataset.

B KEY-BLIND ATTACK AND EVALUATION

Attacker observes protected records, not keys or slot labels; slots are revealed only in a labelled control.



5 Supervised attacker

Single / mean / max MLP: $d \rightarrow 256 \rightarrow 512$
DeepSets: encode, mean, decode
Source embeddings: train targets only



6 Reconstructed embedding

L2-normalized prediction in 512-D
Train on non-test identities
Select checkpoint by validation loss



7 Held-out gallery linkage

Cosine scores vs. unprotected gallery
Top-1 / top-5 / AUROC / EER / TAR
Identity-clustered bootstrap intervals

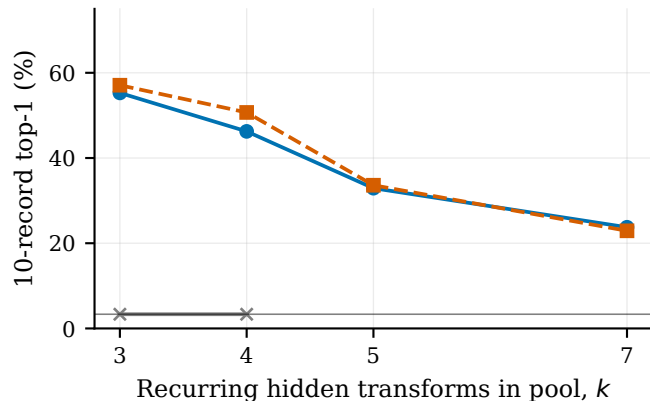
KEY DESIGN Fresh: source-record keys are split-disjoint. Pool K: hidden transforms recur across identity splits.

UNCERTAINTY Earlier studies: 3 model seeds. IoM-GRP / PolyProtect pilots: 1 seed; not confirmation.

SCOPE Tests transform reuse and same-identity aggregation, not a new backbone or a universal privacy guarantee.

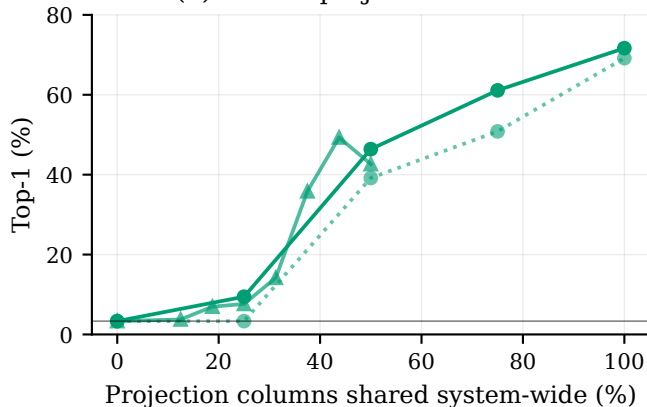
MOBIO mechanism controls (30 gallery identities; chance 3.33%)

(a) Slot label vs. shuffled records



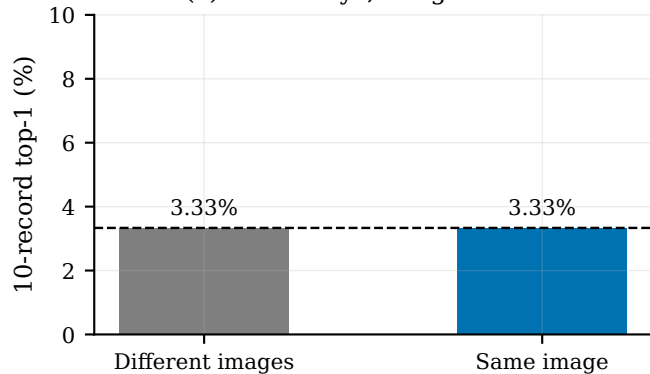
- slot hidden (DeepSets)
- slot label given to attacker
- records 2-10 from other identities

(b) Shared projection columns

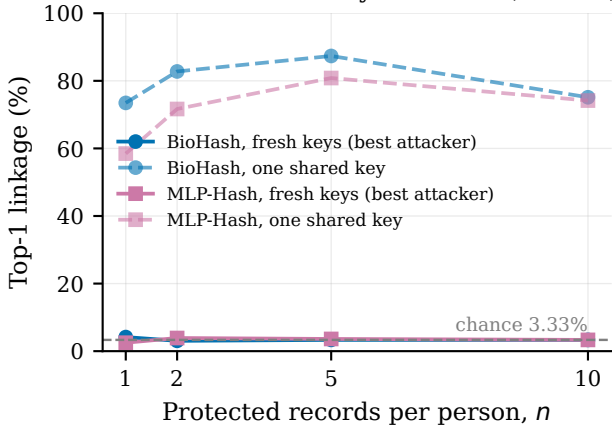


- 10 records, coarse sweep
- 10 records, fine sweep (new split)
- 1 record, coarse

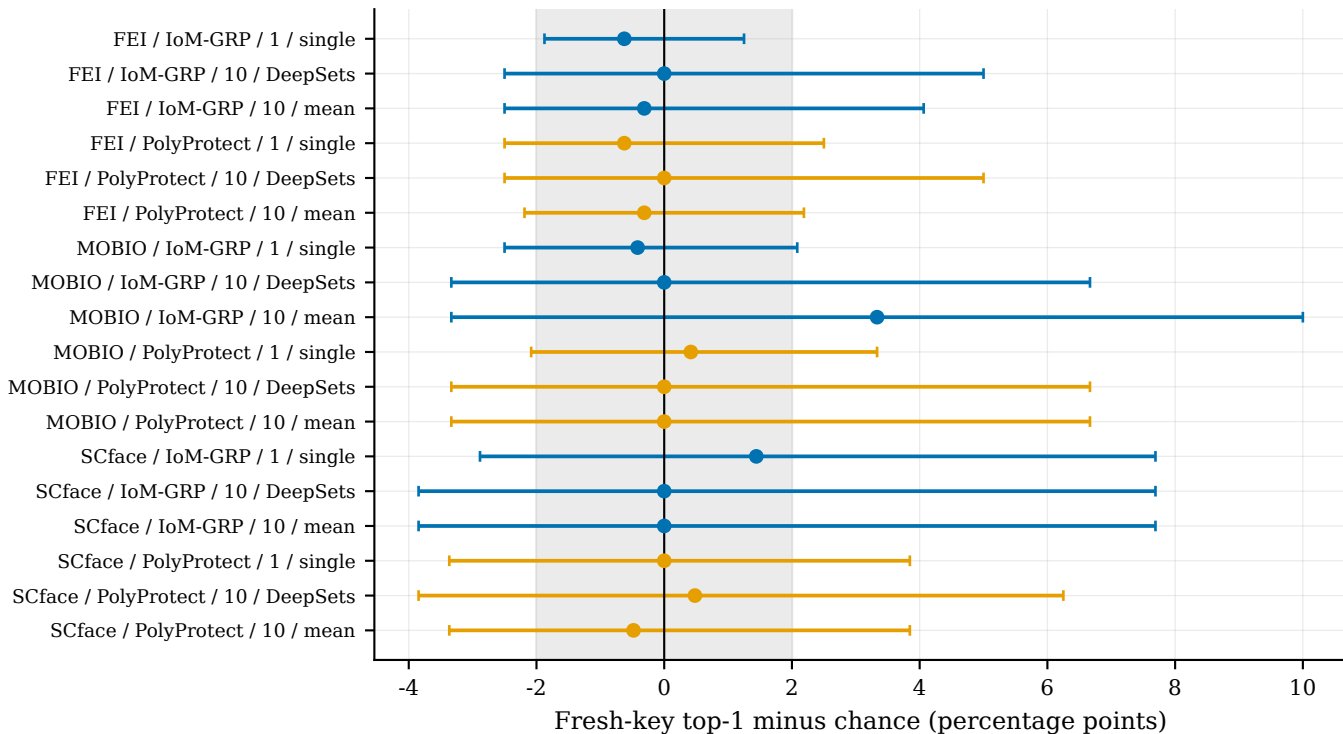
(c) Fresh keys, image control



Fresh and shared keys: MOBIO (N = 30)

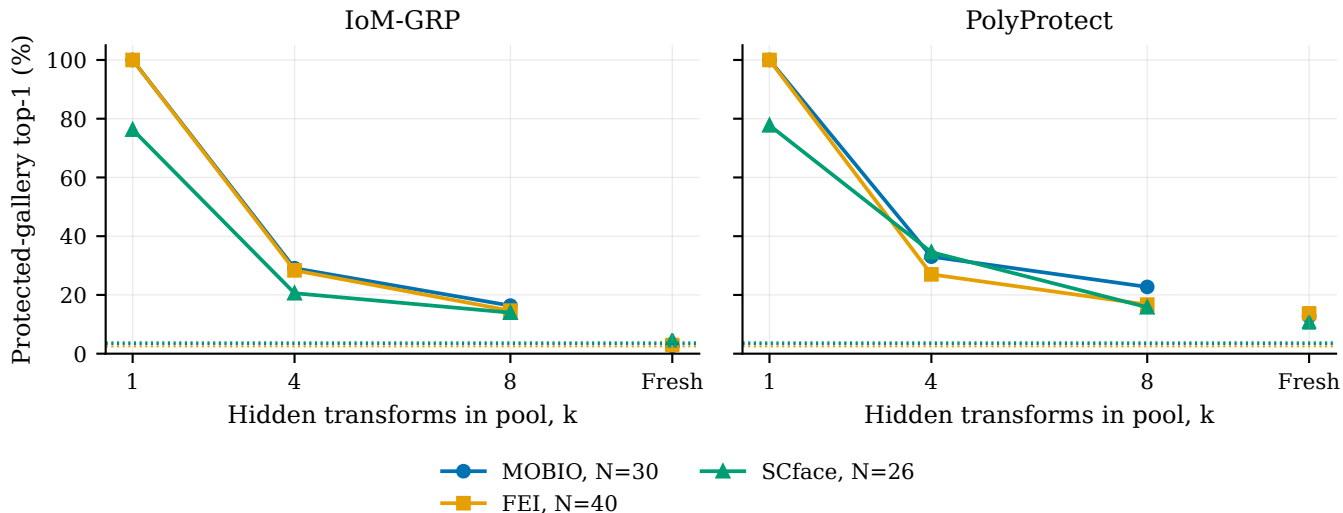


Fresh-key 90% intervals: exploratory pilot sensitivity

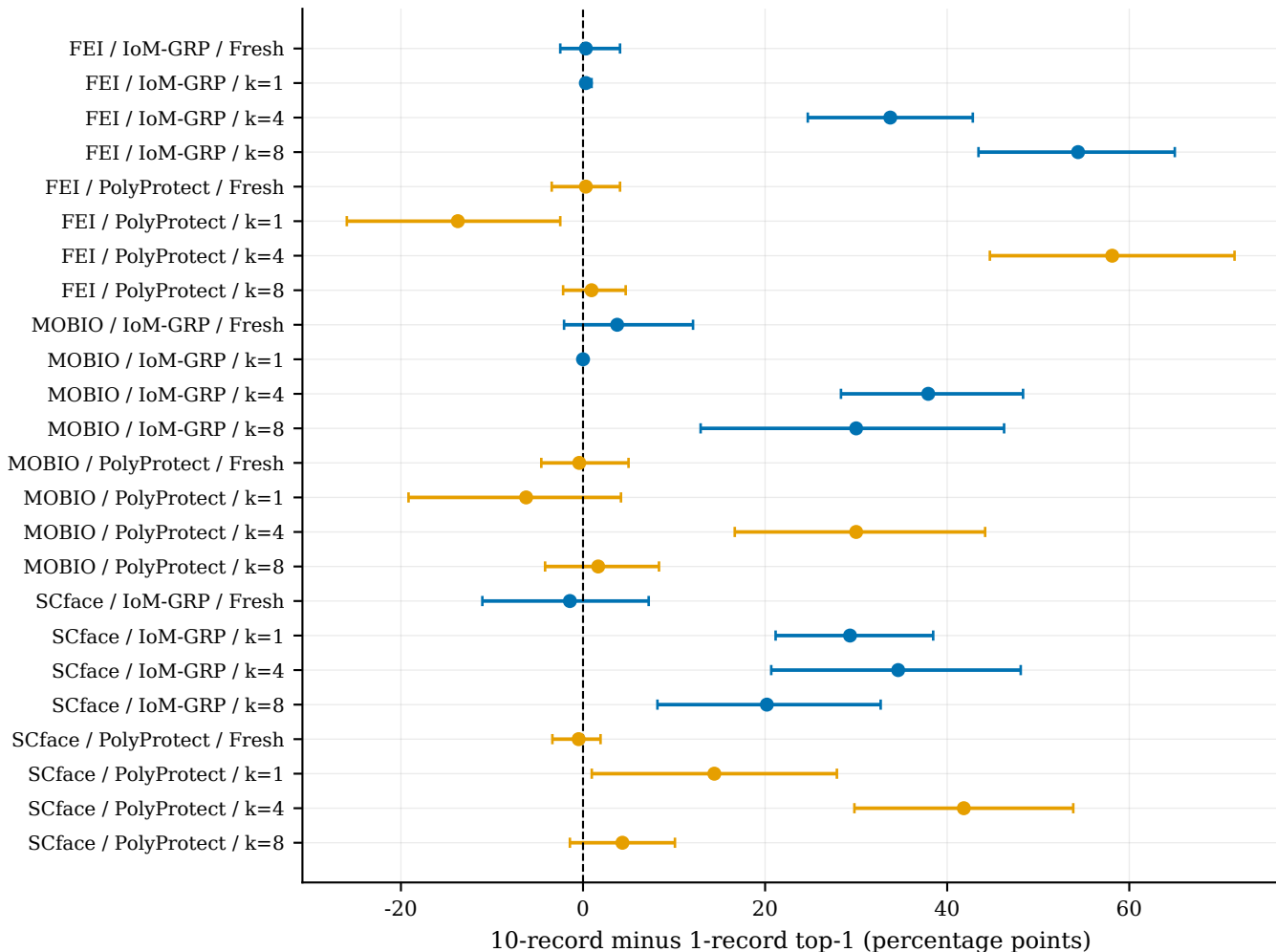


Illustrative +/-2 pp band; not an approved equivalence margin

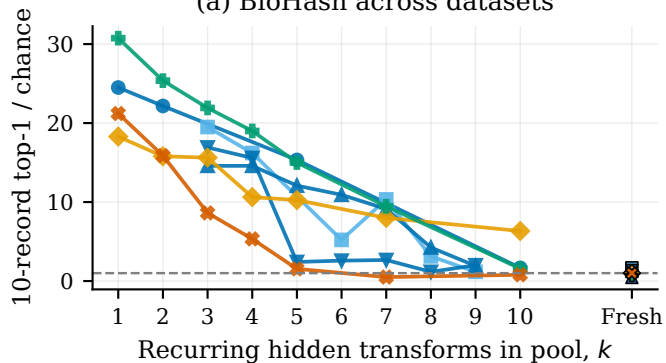
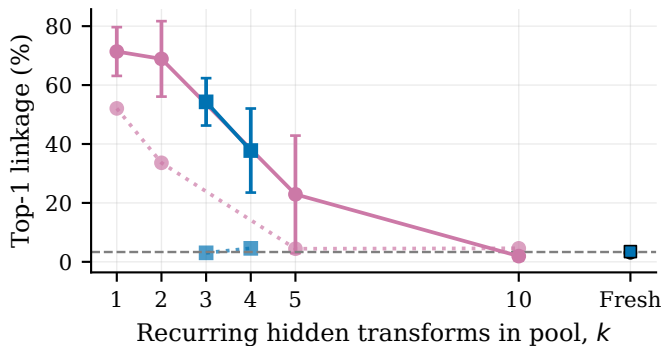
Native matching diagnostic, not learned embedding linkage



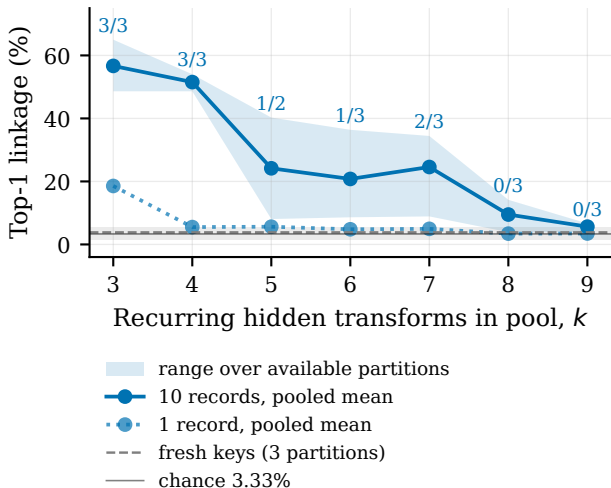
Paired mean-pool gains: one-seed pilots, 95% intervals



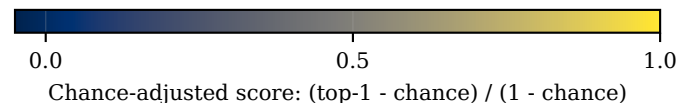
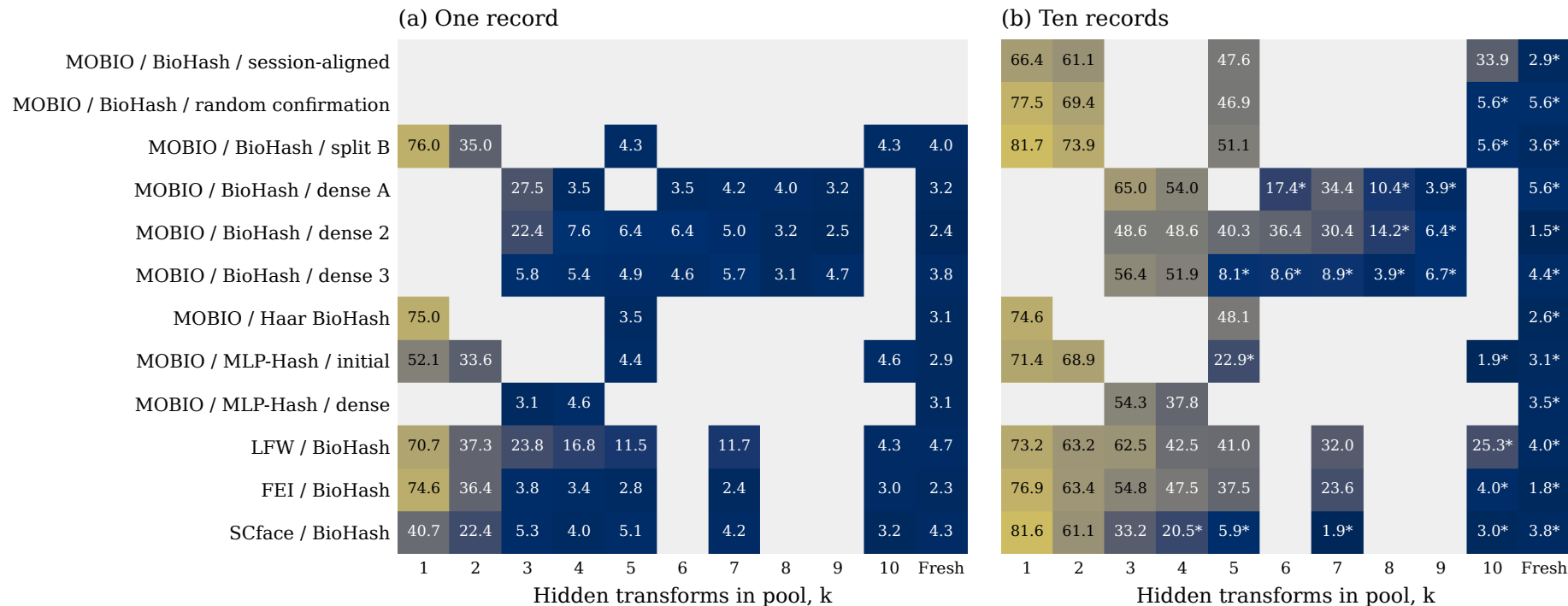
(a) BioHash across datasets

(b) MLP-Hash, MOBIO ($N = 30$)

MOBIO boundary (N = 30)



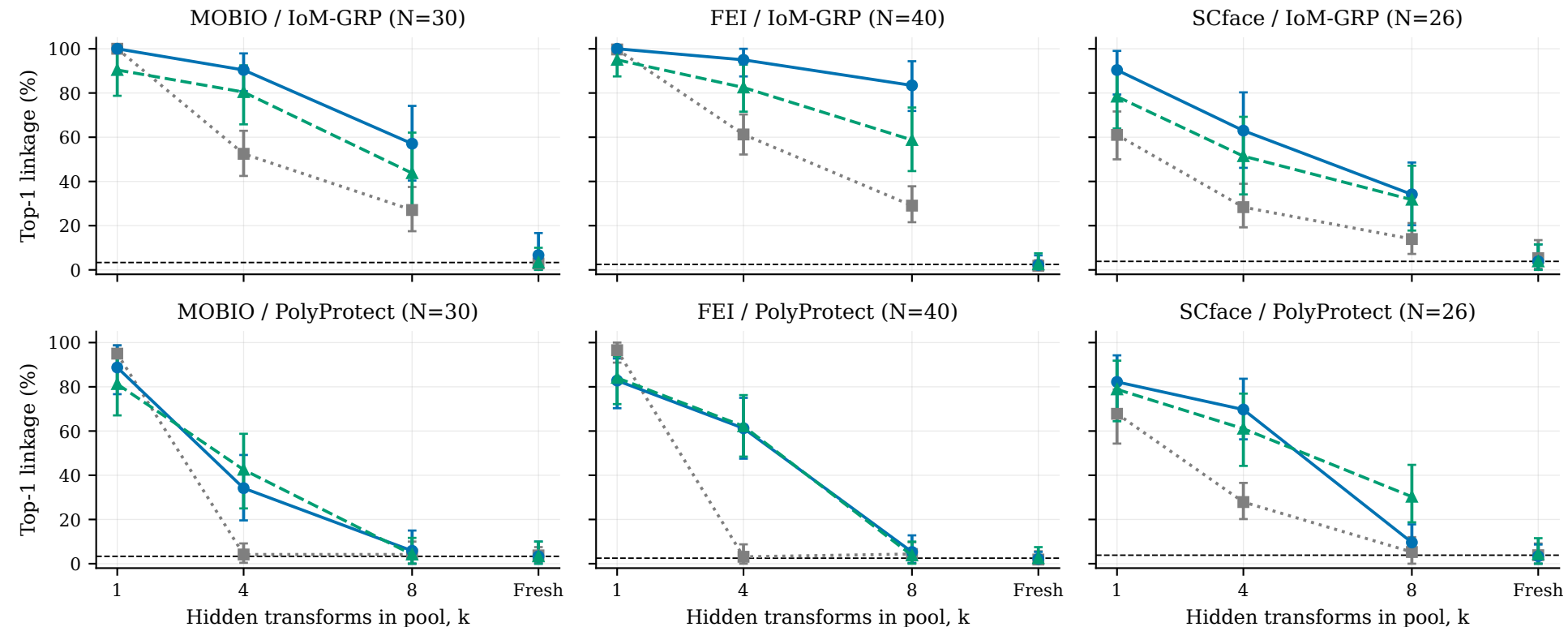
Earlier three-seed key-pool studies



Cells: top-1 (%), 3-seed means. Grey: unavailable. *: not all seed intervals exclude chance.
 Gallery sizes: MOBIO 30; LFW 25; FEI 40; SCface 26. Rows are separate studies, not matched replications.

New protection schemes: one-seed pilots

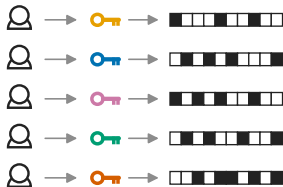
120-epoch cap; bars: identity-clustered 95% intervals; dashed baseline: chance



1 record 10 records, mean 10 records, DeepSets

Fresh keys

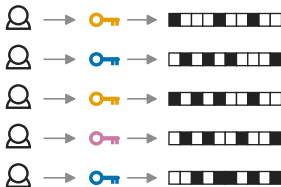
one key per source record



no transform reuse

Recurring key pool

k transforms; slot labels hidden

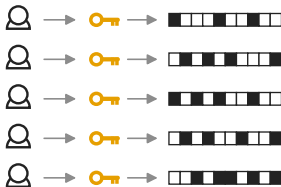


same colour = same transform

reuse across train and test

Single shared key

k = 1; key value still hidden



shared-transform baseline

Learned attacks inconclusive

PolyProtect outside the theorem

Multi-record gains observed

boundary depends on dataset and split

Strong linkage observed

hidden does not mean unlearnable

Protocol

Disjoint train / validation / test identities; one held-out gallery image per identity
8 sets per identity; 3 seeds earlier, 1 in pilots; identity-clustered intervals